



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 7 • STANDARD 6.AT.11

Graph Inequalities

Lesson 7-5 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can graph the solutions of an inequality on a number line.

Language Objective: I can explain my graph using the words number line, open circle, closed circle, and solution set.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Number line

Open circle

Closed circle

Solution set

Inequality

Boundary point



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1 A line with numbers in order used to show values and solutions is a

.

2 A hollow circle that shows a value is NOT included, used for $<$ or $>$, is an

.

3 A filled-in circle that shows a value IS included, used for $\leq$ or $\geq$, is a

.

4 All the values that make an inequality true form the .

5 A math sentence comparing two amounts with $<$, $>$, $\leq$, or $\geq$ is an

.

6

The number where the solution set begins on the number line is the

Reflect — Exit Ticket

How would you graph the inequality $x \geq 9$?

- A. Closed circle at 9, shaded right
- B. Open circle at 9, shaded right
- C. Closed circle at 9, shaded left
- D. Open circle at 9, shaded left

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Number line
2. **Notes 2:** Open circle
3. **Notes 3:** Closed circle
4. **Notes 4:** Solution set
5. **Notes 5:** Inequality
6. **Notes 6:** Boundary point
7. **Watch for this mistake:** *A common mistake in Graph Inequalities is skipping the key idea: "Closed circle for $\leq$ or $\geq$ (included), open circle for $<$ or $>$ (not included); shade right for greater, left for less." — always check your work against this rule before you submit.*
8. **Try It 1:** A. Open circle at 15, shade right — 'Greater than' ($>$) is a strict symbol, so 15 is NOT included — open circle. Larger values work, so shade right.
9. **Try It 2:** A. Closed circle at 4, shade left — 'Less than or equal to' ($\leq$) includes 4 — closed circle. Smaller values work, so shade left.
10. **Exit Ticket:** A. Closed circle at 9, shaded right — $x \geq 9$ means 9 is included (closed circle) and values are 9 or greater (shaded right).